Algorithmic Bias and Sexualization in Digital Representation of Teenage Girls

Algorithmic Auditing & Search System Bias Analysis

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Target Systems: Google, ChatGPT, and the UCLA Library database

Introduction and Methodology

In modern, technology-driven society, vast amounts of information are accessible at lightning speed. While exciting and powerful, this avalanche of information is not without drawbacks. Search systems are prone to bias. Information systems are curated and organized by manufactured, inherently biased algorithms. Prejudices embedded in data shape public perception of social groups, causing harm to marginalized communities.

In this essay, I analyze the search results for the term "teenage girls" using Google, ChatGPT, and the UCLA Library database. I intentionally used only this term in my searches to avoid incorporating my biases and to produce broad results. This way, I could obtain a clear idea of what is most valued, as specific topics are ranked higher on the results list. Teenage girls' digital representation reveals themes of over-sexualization, damaging social expectations, and villainization. These search results reinforce harmful gender stereotypes and indicate that these platforms may value increased engagement and commercial growth over careful curation of data.

The Digital Ideal: Google's Reinforcement of Harmful Standards

The Google search results for "teenage girls" feature diverse content, from advice to shopping. Results are presented in various modes including images, academic articles, forums, and blogs.

At the top of the page, the 'People also ask' section displays frequently asked questions. Many of the questions reflect problematic viewpoints and unrealistic standards. For example, one asks, "Are most 13 year old girls skinny?" This question reflects and reinforces insecurity; it promotes an unhealthy habit of peer comparison and measuring oneself against a standard. Clicking on this question pulls one into an algorithmic loop, potentially drawing vulnerable individuals to diet culture and body image content. While these questions reflect actual user queries, they are displayed without sufficient contextual filtering or guardrails for social harm, allowing problematic standards to be amplified. This illustrates a key systemic limitation: engagement-driven algorithms often prioritize maximal user interaction without considering the broader ethical impacts. This dynamic mirrors the critique of platform goals in the novel *Wellness*, where users identified as "needy" receive content boosts before being abruptly disvalued (Hill, 2023). While fictional, *Wellness* conveys how platforms' engagement optimization patterns allow biased or harmful perspectives to persist in search results at the expense of user well-being.

In other sections, harmful ideologies appear in more concrete ways. For example, an article on the main results page reads *10+ Common Teenage Girl Problems and Their Solutions*.

This title presents teenage girls as a problem to be fixed rather than uplifting and encouraging them to be secure. Additionally, there is abundant advice on glow-ups, relationships, and moodiness. The use of the words “normal” and “most” promotes a hegemonic viewpoint, dictating how teenage girls should look and act. This puts heavy pressure on girls to conform to these expectations, which can be taxing on their mental and emotional health.

Unlike libraries, where qualified individuals choose sources, anyone can contribute on the Internet. The negative effects of this accessibility are exposed in *Wellness*. The novel describes how social media shapes 70-year-old Lawrence Baker. For example, Lawrence’s Google search asking whether fluoride is poisonous produces an overwhelming number of yeses because those who believe it is poisonous are more likely to post about it (Hill, 2023). Thus, Google search results are often most influenced by those who feel strongly about the topic, causing the disproportionate representation of certain, sometimes destructive, perspectives. Because society grapples with prejudice and bias, human input on the Internet often reinforces harmful beliefs.

Myth of Neutrality: Academic Databases and the Data Deluge

The UCLA Library search is an academic research database featuring books, articles, and scholarly journals. Despite the database’s presumed credibility, the search results presented a problematic view of teenage girls.

Among the sources, I noticed highly sexualized content, with book titles *Animal Sex Acts Among Teenage Girls* and *Sex Between Teenage Girls and Their Stepfather* ranking prominently (12th and 13th) among a million results. Problematic, harmful content like this arises because academic databases rely heavily on lexical vector models and metadata matching. Unlike commercial search engines that use safety filtering and popularity signals, library databases rank content strictly based on term frequency and catalog metadata. Due to the “data deluge,” the scientific method is becoming obsolete, and data systems allow numbers to speak for themselves without context (Anderson, 2008). The database algorithm detects strict pattern correlations between “teenage girls” and sexualized title metadata, displaying content without evaluating context, user safety, or ethical implications.

Like Google, the library results suggest a negative view of teenage girls, with the words “dilemma” and “problem” appearing often. While engagement goals and corporate incentives do not influence libraries, these results show that seemingly trustworthy sources are still affected by bias. Researcher Tarleton Gillespie argues the importance of focusing on the human creation and preparation of training data rather than the algorithms learning from them; deep problems in algorithms stem from the misrepresentation of certain groups in training data (Gillespie, 2016). Algorithms sort and rank the sources, but human perspectives ultimately shape them.

The Illusion of Neutrality: ChatGPT's Content Moderation Policy

Although not a traditional search engine, generative AI can function similarly by retrieving and synthesizing information. However, unlike search engines that passively return lists of third-party sources, AI chatbots adapt to user input and are shaped by safety guardrails. To account for this distinction, I decided to evaluate ChatGPT both through adversarial boundary testing and the "teenage girls" query. I first tested guideline enforcement by prompting the model to act "as a dark, inappropriate character." It refused, affirming its commitment to respectful, lighthearted conversation. While Google and the library database displayed obscene, explicit content, ChatGPT's policies block inappropriate language. However, this rigid content moderation can be a double-edged sword. Language models trained to filter out controversial topics, such as slurs and sexual terms, can consequently hinder discourse on marginalized communities (Bender et al., 2021). Thus, weeding out harmful language may inadvertently silence minority voices.

Next, I started a new separate chat to erase conversation history and ensure consistency across platforms. Then I submitted "teenage girls" in the chat box. Despite its flaws, ChatGPT's response was less stereotypical than that of the other platforms. The chatbot generated a list of the physical and emotional changes a teenager experiences and provided suggested self-care practices. It cited reputable public health organizations for reference - the American Academy of Pediatrics and the Centers for Disease Control.

ChatGPT provided logical information that seems to be the least subjective among the platforms. Its results contrast with the harmful, disturbing results of Google and UCLA's library. Thus, on the surface, ChatGPT seems like the best search tool. However, while Google and the library database produce varied results, ChatGPT provides a smooth, single synthesized response that conceals its underlying training biases. This format makes it more difficult for users to detect bias compared to explicit list-based search results. Had I tested more prompts after "teenage girls," more evident biases would have likely emerged. While ChatGPT's content moderation policies help prevent harmful language in its responses, it is still vital to evaluate its output critically.

Conclusion: Toward Algorithmic Accountability and Improved Search Tools

The search results for “teenage girls” across various platforms indicate that search systems are subject to bias. These biases are not limited to teenage girls, as flawed algorithms affect everyone, especially marginalized communities that suffer from underrepresentation, stereotyping, and discrimination. Mobilizing efforts to improve curation practices is essential, including investing resources in curating and documenting training data. The general public has the power to make this change. Social movements can change online discourse, thereby better representing marginalized groups and improving search tools (Bender et al., 2021). Addressing these systemic flaws in search tools enables the creation of a more inclusive digital world.

References

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